

Science—Biology: Photosynthesis

Subject: Science—Biology

Photosynthesis is the process by which green plants, algae, and some bacteria use light energy to make chemical energy. In plants, it mainly occurs in chloroplasts, organelles containing the pigment chlorophyll. The overall equation is commonly written as $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Carbon dioxide enters leaves through small openings called stomata, water is transported from roots through xylem, and light is absorbed by pigments. Glucose represents stored chemical energy, while oxygen is released as a product of splitting water and the later reactions of the process.

Photosynthesis has two connected stages. The light-dependent reactions occur in the thylakoid membranes of chloroplasts. Chlorophyll absorbs photons, and this energy drives the movement of electrons through an electron transport chain. Water is split, producing oxygen, hydrogen ions, and electrons. The reactions form ATP and NADPH, molecules that carry energy and reducing power to the next stage. The light-dependent reactions therefore convert light energy into chemical forms that can be used to build carbohydrates.

The Calvin cycle takes place in the stroma of the chloroplast. It does not use light directly at each step, but it depends on ATP and NADPH made by the light-dependent reactions. Carbon dioxide is fixed into an organic compound, and a series of reactions produces a three-carbon sugar. Some of this sugar is used to regenerate the molecule that accepts carbon dioxide, while some leaves the cycle and contributes to glucose and other carbohydrates. The cycle is sometimes called the light-independent stage, but it generally operates when light reactions are providing energy.

The rate of photosynthesis is affected by light intensity, carbon-dioxide concentration, temperature, water availability, and the condition of the leaf. Increasing light intensity can increase the rate until another factor becomes limiting. If temperature is too low, enzyme-controlled reactions slow; if it is too high, enzymes may lose their functional shape and stomata may close to reduce water loss. Plants in dry environments may have adaptations such as thick cuticles, reduced leaves, deep roots, or specialized timing of gas exchange.

Photosynthesis is central to life on Earth. It introduces energy into most food webs and converts atmospheric carbon dioxide into organic matter. Herbivores obtain chemical energy by eating plants, and consumers receive energy through food chains. Photosynthesis also influences the carbon and oxygen cycles. Cellular respiration uses oxygen and organic molecules to release usable energy, while photosynthesis stores energy and uses carbon dioxide. Forests, grasslands, crops, algae, and ocean microorganisms all contribute to global carbon movement. Protecting photosynthetic organisms and their habitats supports food production, biodiversity, and climate regulation.